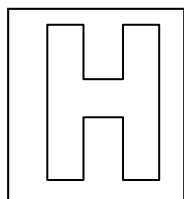


Candidate Name: _____

Class Adm No

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2018 End-of-Year Exams Pre-University 2

H1 Biology

8876/02

Paper 2 Core Paper

10 September 2018

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	/ 11
2	/ 8
3	/ 7
4	/ 10
5	/ 9
Section B	/ 15
Total	/ 60

This question paper consists of 14 printed pages.

[Turn over

Section A

Answer **all** questions in this section.

1. The process of DNA replication is sometimes described as “*asymmetrical*” replication as the daughter strands are synthesised slightly differently in the form of a leading strand and a lagging strand. Fig. 1.1 shows a replication bubble and the origin of replication.

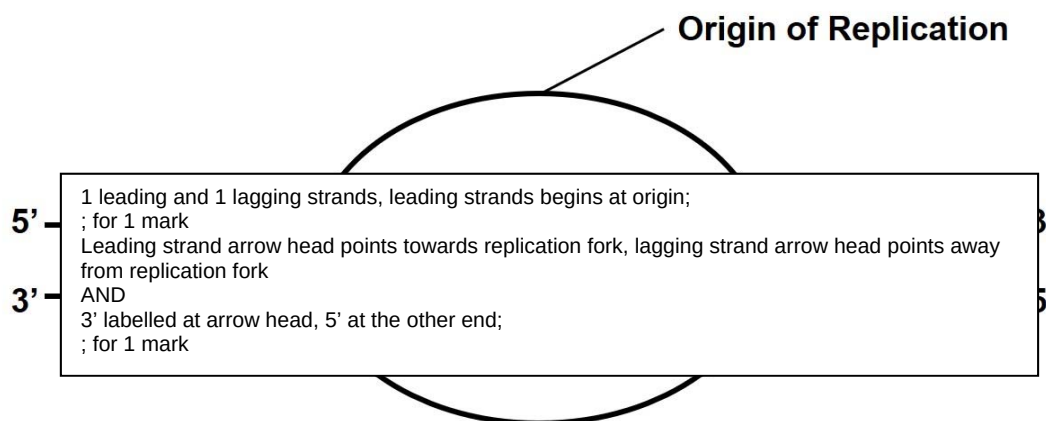


Fig. 1.1

- (a) (i) On Fig. 1.1, draw the following with respect to one of the replication fork:

- Leading and lagging strands
- Arrow heads (→) on leading and lagging strands to indicate direction of synthesis
- 5' and 3' ends of leading and lagging strands

[2]

- (ii) Explain why the two daughter strands are synthesised, “asymmetrically”.

- DNA polymerase can only add new nucleotides to 3' -OH ends;
- DNA is anti-parallel;

; for 1 mark

[2]

Telomeres are short repetitive sequences found at the ends of chromosomes. Length of telomeres can be used to estimate cellular age as telomeres shorten over time.

(b) State what causes telomeres to shorten over time.

End replication problem;
; for 1 mark

[1]

A single human diploid cell contains around six billion base pairs, amounting to around two meters in length. The diameter of a nucleus is around 10^{-15} meters.

(c) Describe how DNA are packaged such that it can fit in the nucleus.

- DNA is wound around histone proteins forming nucleosomes;
; for 1 mark
- Nucleosomes are packed/ fold up to form 30nm chromatin fibre;
- Chromatin fibres are further packed/ fold up to form 300nm loops;
- 300nm loops are further condensed to form chromosomes;
; for 1 mark, max 2 mark

[3]

Senescence cells are cells that have telomeres that have shortened to a critical length such that the cells no longer undergo cell division.

(d) (i) State the cell cycle stage that senescence cells are in.

G₀;
; for 1 mark

[1]

(ii) Explain the purpose for limiting the number of times a cell can divide.

- Prevent the loss of genetic information / coding sequence;
- Prevent the accumulation of mutations;
- Prevent oncogenesis / cancer formation;
; for 1 mark, max 2 marks

[2]

[Total: 11]

2. Rubisco is an enzyme catalysing the first stage of Calvin cycle, its function is vital in maintaining life on Earth.

(a) (i) State the initial reactants and final product(s) of Calvin cycle.

Initial reactants: Carbon dioxide, RuBP, ATP and NADPH
Final Products: G3P, ADP and NADP⁺;
; for 1 mark

[1]

(ii) Explain how Rubisco plays a role in maintaining life on Earth.

- Rubisco catalyses Calvin cycle which converts inorganic carbon to organic carbon;
- Heterotrophs / organisms / secondary consumers that are unable to make their own food relies on plants as food source for organic carbon that is essential for growth/OWTTE;

; for 1 mark

[2]

Rubisco gene is found in all photosynthetic plants. A study was carried out to investigate the effect of temperature on the catalytic activity of Rubisco found in different plants. Fig. 2.1 shows the results.

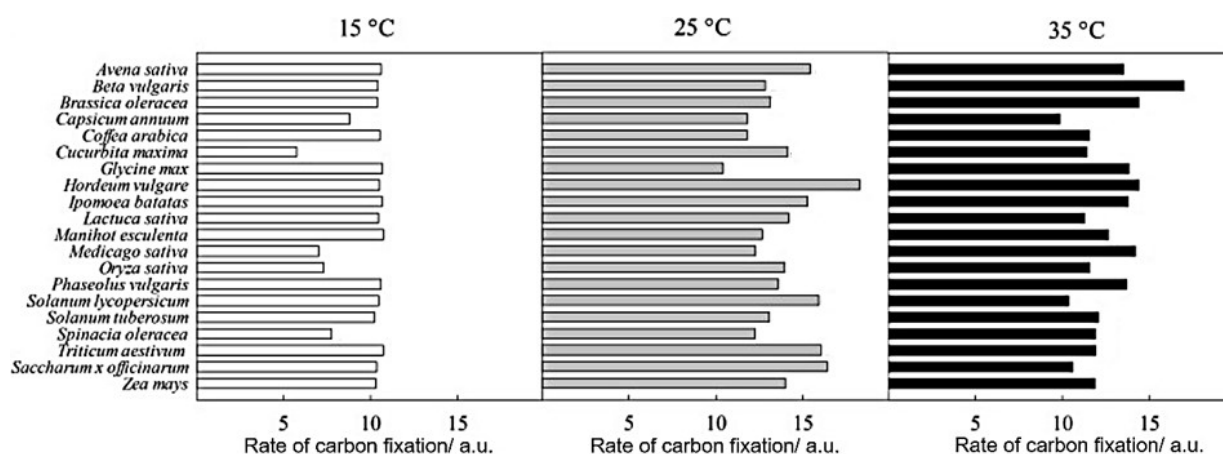


Fig. 2.1

- (b) With reference to Fig. 2.1, conclude if Rubisco found in different plants have the same optimal temperature range.

- Rubisco from different plants have different optimum temperature;
- Rubisco from Beta vulgaris have the highest activity of 17 a.u. at 35°C;
- Whereas Rubisco from Hordeum vulgare have the highest activity of 18 a.u. at 25°C;

(accept any 2 valid comparison with correct name and data set)
; for 1 mark, max 3 marks

[3]

- (c) Suggest why Rubisco from different plants exhibit different catalytic activity under the same temperature condition with all other conditions kept constant.

- Random mutations / genetic variation of Rubisco gene;
- Change in primary structure / sequence of Rubisco results in changes in folding;
- Causes variation in 3D conformation of Rubisco, resulting in variation of catalytic activity;

; for 1 mark, max 2 marks

[2]

[Total: 8]

3. In *Drosophila melanogaster*, the common fruit fly, wing shape is controlled by a gene with multiple alleles. These alleles are listed below in no particular order. When pure-breeding *Drosophila* with long wings were crossed with pure-breeding *Drosophila* with vestigial wings, all offspring had long wings.

Long = W^L

Vestigial = W^V

Antlered = W^a

Drosophila with vestigial wings crossed with *Drosophila* with antlered wings produce all offspring with wings of a new phenotype that appears to be an intermediate between vestigial and antlered phenotype, called vestigial-antlered.

(a) Suggest an explanation for the cross.

- Vestigial allele / W^V and antlered allele / W^a exhibits incomplete dominance;
; for 1 mark
- Both parents are homozygotes / pure-bred / homozygous for vestigial and antlered wings respectively;
; for 1 mark

.....
.....
.....

[2]

Eye colour of *Drosophila* is determined by a gene located on the X chromosome. Sex determination in *Drosophila* is similar to that of humans. *Drosophila* can either have red eyes or white eyes, the allele coding for red eyes is dominant over the allele for white eyes.

Pure breeding female *Drosophila* with long wings and red eyes were crossed with pure breeding male *Drosophila* with vestigial wings and white eyes.

In the F1 generation, all female and male *Drosophila* had long wings and red eyes.

Female flies and male flies from F1 generation were then crossed to produce the F2 generation.

The observed number of F2 flies with each phenotype were as follows:

Female, long wings, red eyes	595
Female, vestigial wings, white eyes	201
Male, long wings, red eyes	304
Male, long wings, white eyes	299
Male, vestigial wings, red eyes	101
Male, vestigial wings, white eyes	100

Draw a genetic diagram to explain the observed results of the cross between female flies and male flies from the F1 generation.

Let X^R be the allele on the X-chromosome that code for red eyes

Let X^r be the allele on the X-chromosome that codes for white eyes

Let W^L be the allele coding for long wings

Let W^V be the allele coding for vestigial wings

F1 phenotype	Female, long wings, red eyes			Male, long wings, red eyes		
F1 genotype	$W^L W^V X^R X^r$			$W^L W^V X^R Y$		
Gamete						
Punnett square	$W^L X^R$	$W^L W^L X^R X^R$ Female, long wings, red eyes	$W^L W^L X^R Y$ Male, long wings, red eyes	$W^L W^V X^R X^R$ Female, long wings, red eyes	$W^L W^V X^R Y$ Male, long wings, red eyes	
	$W^L X^r$	$W^L W^L X^R X^r$ Female, long wings, red eyes	$W^L W^L X^r Y$ Male, long wings, white eyes	$W^L W^V X^R X^r$ Female, long wings, red eyes	$W^L W^V X^r Y$ Male, long wings, white eyes	
	$W^V X^R$	$W^L W^V X^R X^R$ Female, long wings, red eyes	$W^L W^V X^R Y$ Male, long wings, red eyes	$W^V W^V X^R X^R$ Female, vestigial wings, red eyes	$W^V W^V X^R Y$ Male, vestigial wings, red eyes	
	$W^V X^r$	$W^L W^V X^R X^r$ Female, long wings, red eyes	$W^L W^V X^r Y$ Male, long wings, white eyes	$W^V W^V X^R X^r$ Female, vestigial wings, red eyes	$W^V W^V X^r Y$ Male, vestigial wings, white eyes	
F2 genotype	$1/16 W^L W^L X^R X^R$ $1/16 W^L W^L X^R X^r$ $2/16 W^L W^V X^R X^R$ $2/16 W^L W^V X^R X^r$	$1/16 W^V W^V X^R X^R$ $1/16 W^V W^V X^R X^r$	$1/16 W^L W^L X^R Y$ $2/16 W^L W^V X^R Y$	$1/16 W^L W^L X^r Y$ $2/16 W^L W^V X^r Y$	$1/16 W^V W^V X^R Y$ $1/16 W^V W^V X^r Y$	
F2 phenotype	Female Long wings, red eyes	Female Vestigial wings, Red eyes	Male Long wings, red eyes	Male Long wings, white eyes	Male vestigial wings, red eyes	Male vestigial wings, white eyes
F2 Phenotypic ratio	6 :	2 :	3 :	3 :	1:	1

Correct parental phenotypes and genotypes;

Correct gametes drawn;

Correct Punnett's square drawn including phenotypes;

Correct F2 phenotypes and genotypes;

Correct phenotypic ratio;

[5]

[Total: 7]

4. Global temperature records show temperature anomaly ($^{\circ}\text{C}$), which is the difference in temperature of the measured year to the average temperature of a reference year-span. Fig 4.1 shows the global temperature records from 1900 to 2014 relative to the average temperature from 1951 to 1980.

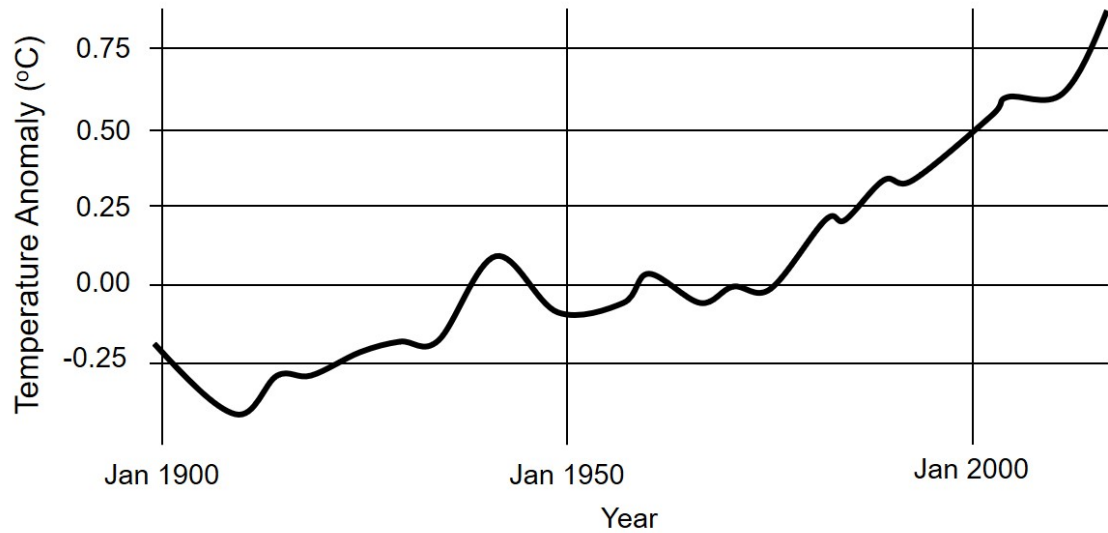


Fig. 4.1

Fig. 4.2 shows the global carbon emission from burning of fossil fuel from 1900 to 2014.

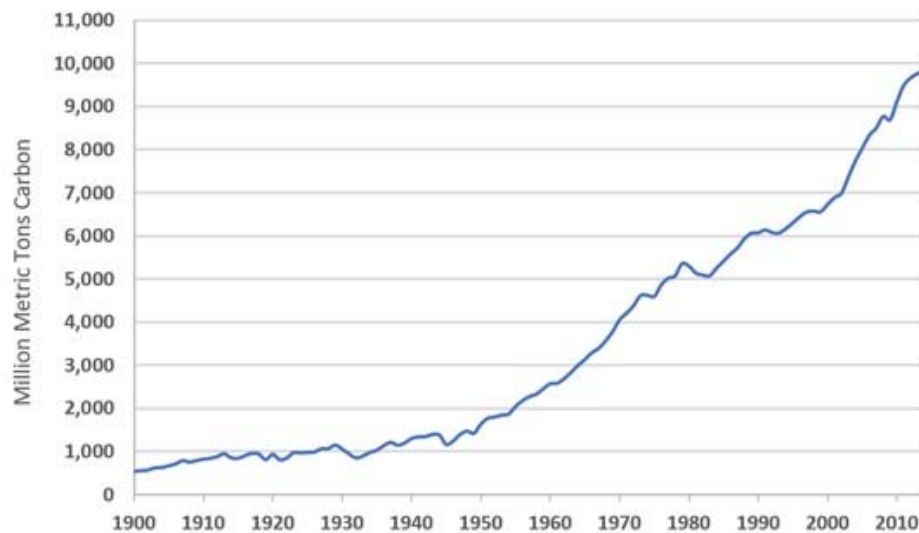


Fig. 4.2

(a) Comment on the trend observed for both global temperature records and global carbon emission from 1950s onwards.

- Temperature anomaly increased from -0.2°C to around +0.8°C while carbon emission increased from around 1500 million metric tons to just below 10000 million metric tons;
- Increase in carbon emission imply the increase in the greenhouse gas carbon dioxide;
- Increase greenhouse gas results in absorbing of more solar radiation / enhancing greenhouse effect causing temperature to rise;
; for 1 mark, max 2 marks

.....
.....
.....

[2]

Changes in global temperature will have an impact on many other abiotic and biotic factors, one of them being sea levels.

(b) With reference to Fig. 4.1, account for how sea levels have been affected.

- Rising temperatures have caused sea levels to rise;
; for 1 mark
- Higher temperature causes melting of ice sheets, which adds fresh water to sea water;
- Higher temperature causes thermal expansion of water;
- Loss of ice sheets leads to less reflective surface, more heat absorbed by sea water which further accelerates melting of ice sheets;
; for 1 mark

.....
.....
.....

[3]

Another factor that is impacted by changes in global temperature is the spread of mosquito borne diseases like Dengue fever (DF), caused by the dengue virus. Fig. 4.3 shows the number of reported DF cases globally from 1980 to 2002.

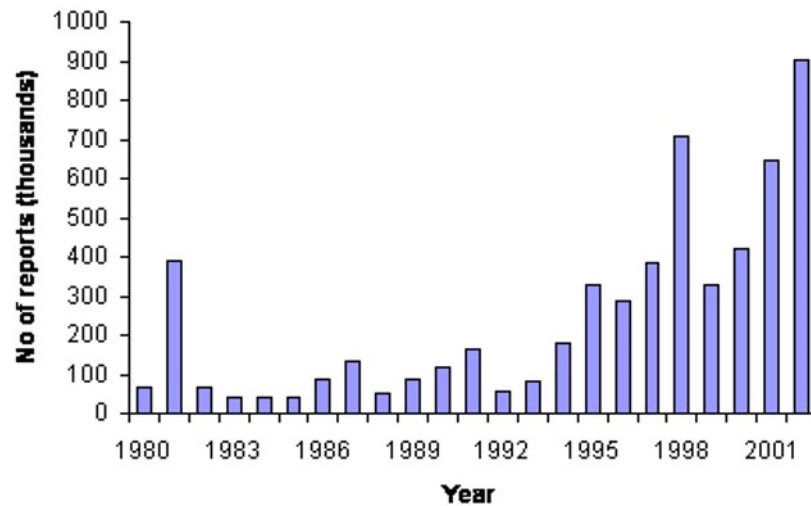


Fig. 4.3

(c) Explain how the data in Fig. 4.1 can account for the general trend observed in Fig. 4.3.

- Increased temperature, increased dengue fever cases;
 - Higher temperature leads to increased survival rate / less prone to predation / parasites and diseases of Aedes Mosquito;
 - Higher temperature leads to increased metabolism rate in Aedes Mosquito;
 - accelerates emergence of mosquitoes /shorter hatching time / life cycle shorten;
 - increasing mosquito population;
 - reduce extrinsic incubation period of virus in insect;
 - increasing the number of infective vector;
 - female mosquitoes digest blood faster and feed more frequently;
 - higher temperature may lead to increased precipitation, resulting in more breeding sites for mosquitoes;
 - Expand towards higher latitudes;
- ; for 1 mark, max 3 marks

[3]

The Intergovernmental Panel on Climate Change (IPCC) has made predictions on future global surface temperature based on past trends and existing data. If no actions are done, the trend observed in Fig. 4.1 can be expected to continue.

(d) Assuming if the trend in Fig. 4.1 continues, predict how dengue would spread in the future.

- | | |
|---|--|
| <ul style="list-style-type: none"> • Global surface temperature is predicted to continue to increase, causing areas <u>outside of the tropics to get warmer</u>; • Mosquitoes would spread to areas outside of the tropics, as such diseases like dengue would also <u>spread beyond the tropics / towards to poles</u>; • Lower altitudes may get too hot, causing mosquitoes to spread to cooler <u>higher altitudes</u>; • AVP | <p>.....</p> <p>.....</p> <p>.....</p> |
|---|--|

; for 1 mark

.....[2]

[Total: 10]

5. The length of the upper leg bone is an important variable in research on human evolution as it provides information about the energy used in movement.

Fig. 3.1 shows the length of upper leg bone that have been measured or estimated from fossils of four hominid species (*Australopithecus*, *H. erectus*, *H. neanderthalensis* and *H. sapiens*) covering a time period of approximately three and a half million years.

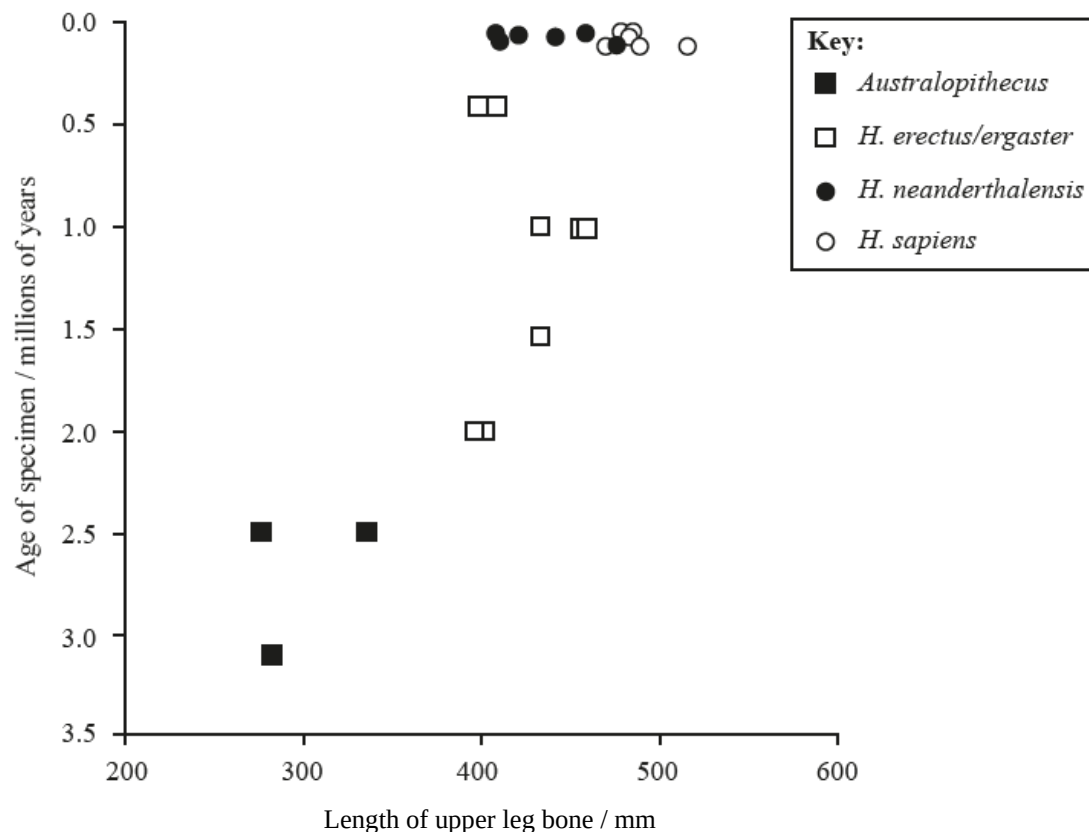


Fig. 3.1

Source: Adapted from K. Steudel-Numbers and M. Tilkens (2004) *Journal of Human Evolution*, 47, pages 95–109.
‘The effect of lower limb length on the energetic cost of locomotion: implications for fossil hominins’

- (a) With reference to Fig. 3.1, explain if increasing upper leg bone length provides an evolutionary advantage.

Overall trend of increasing upper leg bone length from ancient to recent species;
Upper leg bone length increases from about 300 mm in the oldest/most ancient *Australopithecus* to about 500 mm in modern human/*H. sapiens*;

; for 1 mark, max is 1 mark

Longer upper leg bone is a selective advantage + may allow for more energy-efficient/faster movement/upright posture/gait/taller to see predators/other appropriate reason;
Species with shorter upper leg bone are selected against/died out;
However, overlap in ranges for more recent specimens/ *H. neanderthalensis* and *H. sapiens* suggests no strong selective advantage;

for 1 mark, max is 2 marks

...[3]

(b) Suggest one limitation of using fossil records in evolutionary studies.

- | | |
|--|-------------------|
| .. Fossil specimens may not be identified correctly;
.. Age of specimens may not be accurate;
.. Fossils may be incomplete;
.. for 1 mark, max is 1 marks |
.....[1] |
|--|-------------------|

We would expect natural selection to eliminate genetic variations with negative effects from human population. However, the study on human genetics reveals that several recessive alleles that are known to cause diseases have been preserved in human populations.

For instance, the sickle-cell allele is carried by half the people in some areas of Africa. This distribution seems to result from the counterbalancing effects of anaemia and malaria, a disease that formally cause high death rates in equatorial Africa.

(c) Account for the high frequency of sickle cell anaemia allele in the African populations.

- | | |
|---|--|
| <u>Heterozygous advantage:</u>
Heterozygotes do not exhibit sickle cell anaemia / phenotypically normal;
Heterozygous / sickle cell traits individuals are at <u>selective advantage</u> in malaria affected areas / more <u>resistant to malaria</u> ;
Heterozygous individuals survive to reproduce and pass on the sickle cell allele to offspring;
Hence <u>increasing the HbS allele frequencies</u> in the population within these malaria areas over time;
for 1 mark, max is 3 marks |
.....
.....
.....
.....[3] |
|---|--|

(d) Explain why population is the smallest unit that can evolve.

- | | |
|--|-------------------------------------|
| Evolution takes place over <u>extended periods of time/many generations</u> through <u>adaptations/favourable characteristics</u> that are passed through the <u>reproduction</u> between individuals;
Evolution is measured only as <u>change</u> in relative <u>proportions of variations</u> / <u>change in allele frequency</u> in a <u>population</u> over a succession of generations;
<u>Natural selection</u> involves <u>interactions between individual</u> organisms and their <u>environment</u> ;
<u>Variation</u> exists among individuals of a population;
<u>Natural selection</u> favours the <u>survival and reproduction</u> of some individuals over others;
for 1 mark, max is 2 marks |
.....
.....[2]
..... |
|--|-------------------------------------|

tal: 9]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

1.

(a) Compare aerobic and anaerobic respiration in animals. **[7]**

(b) Both Darwin and Wallace independently came up with the theory of evolution by natural selection. They concluded that organisms had to struggle for survival, and that each subsequent generation slowly became better adapted to the environment.

Using named example, explain how environmental factors act as forces of natural selection. **[8]**

[Total: 15]

2.

(a) Contrast between stem cells and cancer cells. **[6]**

(b) Based on data collected over the years on weather patterns and greenhouse gas emissions, researchers have concluded a causal relationship between greenhouse gas emission and enhanced global warming. Biotic factors like coral reefs are sensitive and vulnerable to changes in climate.

Discuss how weather patterns would be affected and the consequences of such changes, along with how corals reefs are impacted by enhanced global warming. **[9]**

[Total: 15]

End of Paper

1.

(a) Compare aerobic and anaerobic respiration in animals.**[7]**

Similarities:

- [Purpose] Both serve to generate energy in terms of ATP molecules to sustain energy-requiring activities;
- [Common process] Glycolysis is common to both, whereby glucose is broken down into pyruvate, giving rise to a net gain of 2 ATP molecules, and producing 2 NADH molecules as well in the cytoplasm;
- [Involvement of hydrogen acceptor] Both needs NAD⁺ as a hydrogen acceptor / co-enzyme for dehydrogenase to complete dehydrogenation reactions in glycolysis;
- [Regeneration of NAD⁺] Both have a process to regenerate the hydrogen acceptor NAD⁺ to support further glycolysis (can also put this as a difference: describe the mechanism involved in regenerating NAD⁺);
- Both only harvest a portion of the chemical energy stored in glucose molecule, with a large amount of energy lost as heat;
- AVP

; for 1 mark, max 5 marks

Differences:

Feature	Aerobic respiration	Anaerobic respiration
Location	In cytoplasm and mitochondria	Only in cytoplasm
Hydrogen acceptor	NAD ⁺ and FAD	Only NAD ⁺
Stages involved in generating ATP	Glycolysis, link reaction, Krebs cycle and oxidative phosphorylation	Only glycolysis and fermentation
Mechanism of ATP synthesis	Direct ATP synthesis via Substrate level phosphorylation during glycolysis and krebs cycle; Oxidative phosphorylation through electron transport chain and chemiosmosis.	Only substrate-level phosphorylation during glycolysis
Number of ATP synthesized per glucose molecule	38 (4 via substrate-level phosphorylation and 34 via oxidative phosphorylation)	Only 2 via substrate-level phosphorylation
Final products	CO ₂ , ATP, H ₂ O	Lactate, ATP
Type of chemical reactions involved in breaking down glucose	Dehydrogenation, Oxidative decarboxylation,	Only dehydrogenation
Mechanism where hydrogen acceptor is regenerated	Via electron transport chain during oxidative phosphorylation, whereby NADH and FADH ₂ donate the electrons and protons to the electron carriers	Via reduction reaction (pyruvate to lactate) during fermentation
Involvement of O ₂	Yes, as the final electron acceptor in the ETC (Reject: just 'Yes' without elaboration)	No.

; for 1 mark, max 5 marks

(b) Using named example, explain how environmental factors act as forces of natural selection.**[8]****Example 1**

- Industrial melanism in Britain/United Kingdom;
- Before the industrial revolution, light-coloured peppered moths dominated the moth population with the dark coloured/melanic moths being largely uncommon;
- Since the light-coloured peppered moths are extremely well camouflaged as its colouration merges with the pale lichens on trees;
- Light-coloured moths can better hide/escape from predators and are selected for to survive and reproduced;
- However, after industrial revolution, it was recorded that most of the moth population were dark coloured/melanic moths;

; for 1 mark, max is 3 marks

- During the industrial revolution, pollution from the burning coal killed off the lichens growing on trees in industrial areas, exposing the darker bark, which was further darkened by soot deposits;
- Therefore, light-coloured peppered moths are no longer as well camouflaged, and it became very conspicuous/obvious to its predators (name one predator) when they rest on the trunks of trees;
- OR
- The dark coloured/melanic form is in turn, now well camouflaged against the dark tree barks;
- This result in the predators (name one example), which acts as the selection pressure, select against the light-coloured peppered moth;

- It reduces the frequency/number of light-coloured peppered moth but increases the dark coloured/melanic form in polluted areas and vice versa for non-polluted areas/OWTTE;
- ; for 1 mark, max is 4 marks

Example 2

- The finches living on the Galapagos Islands;
 - Each island had its own kind of finch (14 in all), found nowhere else in the world/OWTTE;
 - The finches are closely related species arising from a common ancestor;
 - Adaptive radiation led to the formation of so many species;
- ; for 1 mark, max is 2 marks
- Each of which specialises in eating a different type of food/diet;
 - There were 3 main groups, ground, warbler, and tree finches, classified according to their beak sizes;
 - This diversification into different ecological niches, is thought to be necessary to permit the coexistence of different species;
 - In different parts of the Galapagos Islands, different types of beak sizes dominate/OWTTE;
 - In this case, the different types of food available act as the selection pressure;
 - Selecting for a particular beak size that favours adaptation and survival in a particular environment with a certain available type of food;
- ; for 1 mark, max is 3 marks
- *Name two examples of beaks:*
 - Some had large and heavy beaks adapted for eating large seeds, others for small seeds;
 - Some had parrot-like beaks for feeding on buds and fruits;
 - Some had long and slender/thin beaks for feeding on small insects;
- ; for 1 mark, max is 2 marks

QWC:

Student's chosen example correctly identifies the selection pressure, selective advantage / disadvantage and outcome of natural selection

2.

(a) Contrast between stem cells and cancer cells.

[6]

1. Unlike stem cells, cancer cells do not differentiate;
 2. Cancer cells divide indefinitely/ uncontrollably / do not respond to anti-growth signals while stem cell division is determined by molecular signals [that either stimulates cell division or stop it];
 3. Cancer cells have self-sufficiency of growth signals, whereas stem cells relies on external growth signals;
 4. Cancer cells experience no contact inhibition and is invasive while stem cells experience contact inhibition;
 5. Cancer cells have abnormal morphology, whereas stem cells have more regular morphology;
 6. Cancer cells metastasize (dislodge from original tumour and form secondary tumours) while stem cells remain in tissue of origin;
 7. Cancer cells are able to signal for angiogenesis while stem cells do not;
 8. Cancer cells have accumulated loss of function mutation in tumour suppressor genes and gain of function mutations converting proto-oncogenes to oncogenes;
 9. Genome of cancer cells are unstable, there is high tendency of genome alteration / mutations where as stem cells have relatively more stable genomes, with mutations occurring at a much lower rate;
 10. Cancer cells expresses high levels of telomerase activity whereas only zygotic/totipotent/embryonic/pluripotent expresses high levels of telomerase;
 11. Stem cells responds to signal for programmed cell death / apoptosis, whereas cancer cells do not respond / avoids programmed cell death / apoptosis;
AVP;
- ; for 1 mark

(b) Based on data collected over the years on weather patterns and greenhouse gas emissions, researchers have concluded a causal relationship between greenhouse gas emission and enhanced global warming. Biotic factors like coral reefs are sensitive and vulnerable to changes in climate.

Discuss how weather patterns would be affected and the consequences of such changes, along with how corals reefs are impacted by enhanced global warming.

[9]

Weather patterns

1. A warmer climate creates an atmosphere that can collect / retain / drop more water, changing weather patterns;
2. This causes wet areas to become wetter / dry areas become drier;
3. This can lead to heat waves / heavy rain;
4. Heatwaves associated with low humidity may result in wildfire;
5. Heavy rain increases the amount of runoff into rivers and lakes, washing sediment, nutrients, pollutants, trash, animal waste, and other materials into water supplies (any 2), making them unusable/ unsafe/ need of water treatment;
6. Increase rainfall leads to increase breeding sites for mosquitoes, which would lead to an increase in mosquito-borne diseases like dengue;

; for 1 mark, min 2 marks

Coral reefs

7. Heat stress can cause coral bleaching;
8. This is because at higher temperatures, zooxanthellae photosynthesis is disrupted / zooxanthellae produce more toxic compounds;
9. Toxic compounds damages metabolism of coral poly which expels the zooxanthellae, leaving the coral skeleton a stark, "bleached" white;
10. Zooxanthellae and corals have a symbiotic relationship, where zooxanthellae provide corals with food;
11. If temperatures remain above the bleaching threshold for prolonged periods of time, corals will eventually die from starvation and disease /OWTTE;
12. Increase in atmospheric CO₂ cause more CO₂ to be dissolved in the ocean, decreasing the pH./ resulting in ocean acidification;
13. Ocean acidification affects hard corals as they cannot absorb the calcium carbonate to maintain their skeletons / the stony skeletons that support corals will dissolve;

; for 1 mark, max 6 marks

QWC:

Student's answer cover at least 2 points from each section.

[Total: 15]